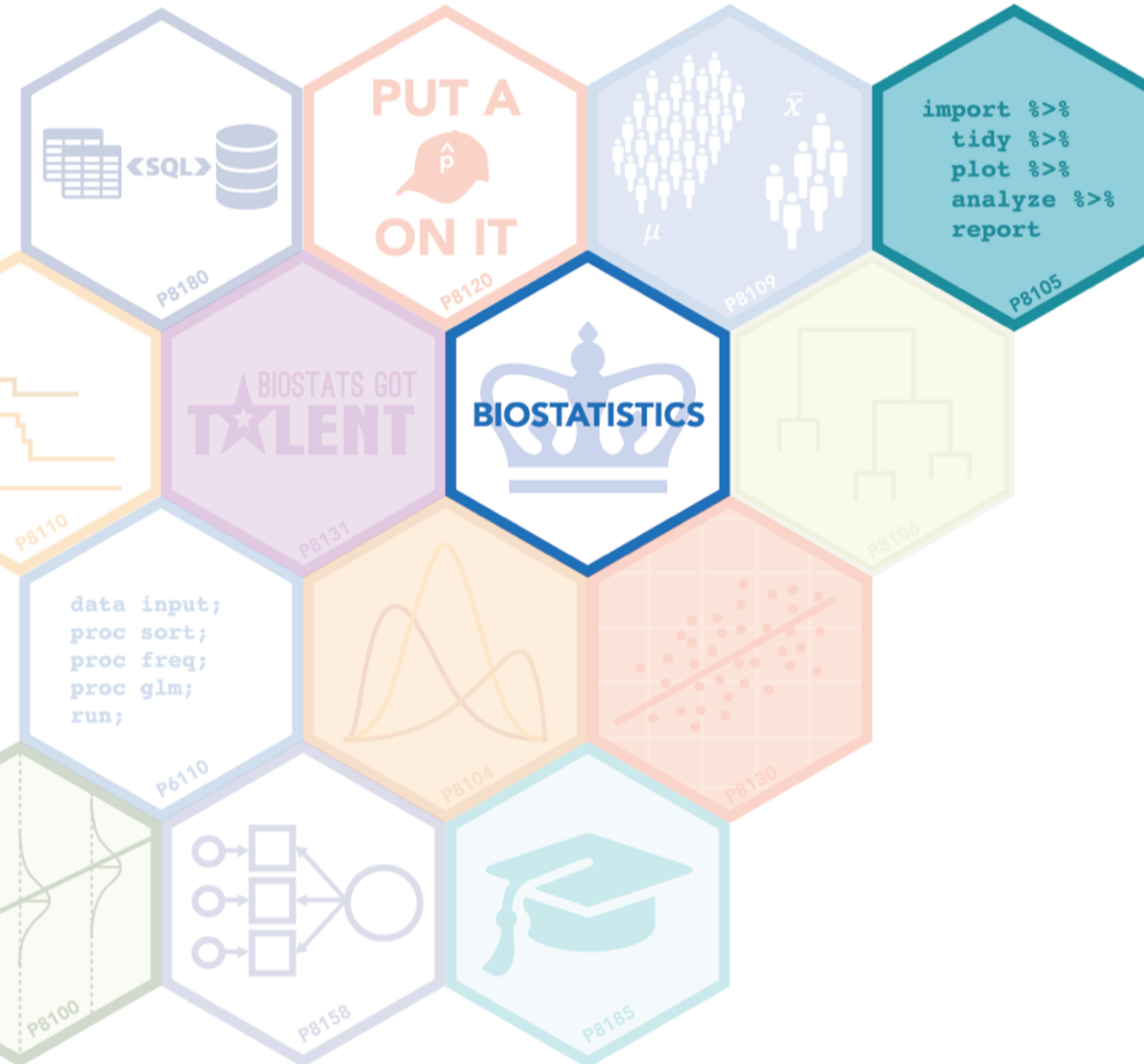




WRITING WITH DATA

Jeff Goldsmith, PhD
Department of Biostatistics

Writing is important

- You're going to spend a lot of your time communicating in writing
 - With collaborators, a general public, future you
 - About data cleaning, analyses, results
 - In formal reports, brief summaries, replies to questions
- Time to get good

Tools

- Code is necessary but not sufficient
- Use tools that combine your code and text
- Greatly facilitates *reproducibility*, which is a big concept
 - In short, someone you don't know or work with should be able to reproduce each step of your analysis
 - As a part of this, they should understand why you did what you did
 - (Again, this someone is often future you)
- We'll use R Markdown to write reproducible reports

General tips

- Know your audience
 - Are they statistically knowledgeable?
 - How many details do they want / need?
- Say exactly what you did
 - Don't leave any thing important out
 - Not the same as a step-by-step list of what you typed into R

General structure

- Introduction / overview
 - Data and methods
 - Data sources
 - Summary statistics
 - Exploratory analysis
 - Formal analysis
 - Results
 - Discussion
-
- Some version of these exist in almost everything I write
 - Sometimes these are long, sometimes they're a sentence

Introduction

- What is the context for this problem?
- What kind of data were gathered?
- What do you hope to learn?

Data

- Importing, tidying, and editing
 - Loading data
 - Reorganizing into usable form
 - Identifying missing values
 - Recoding and creating variables
- Summary statistics
 - Sample size
 - Means or proportions of major variables

Methods / models

- Exploratory analyses
 - Visualizations
 - Numerical summaries
- Formal analyses
 - Model components
 - Model strategy
 - Formal comparisons of interest, tests, significance levels

Results

- What did you find in exploratory analyses (any missing values? data distributions? notable features?)
- What happened in your modeling?
- What is your final model, and what are the important quantities?

Discussion

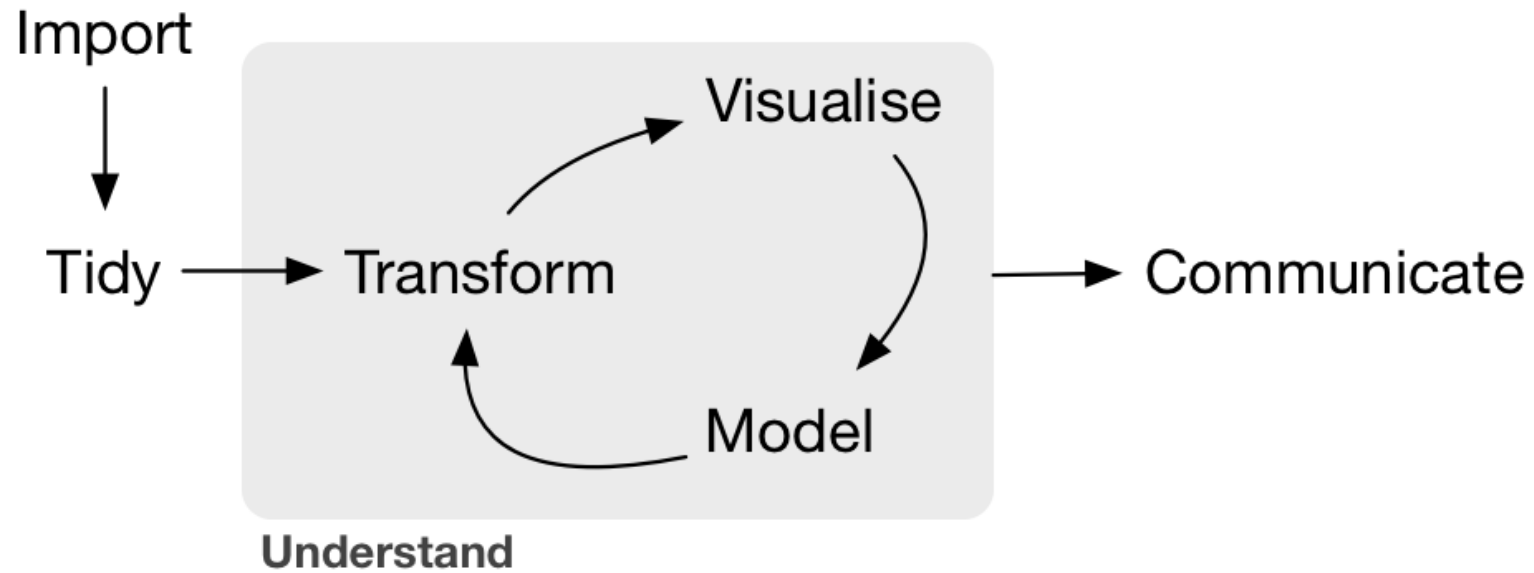
- What do your results say about the question you hoped to answer?
- What were the limitations of your data or your analysis?
- What open questions remain? Are any of these solvable with the current data?
- What are your next steps?

Some true stuff about writing

- It is not easy
- It takes practice
- It is critical to do well

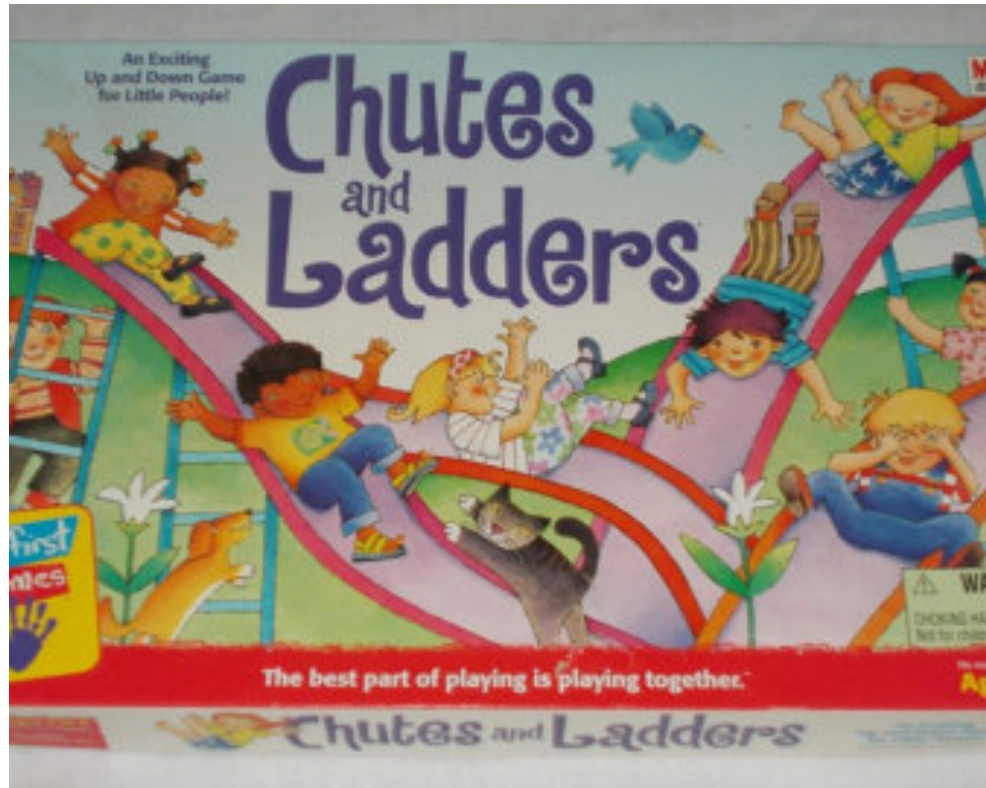
- (LLMs don't "fix" any of that)

A data analysis flowchart

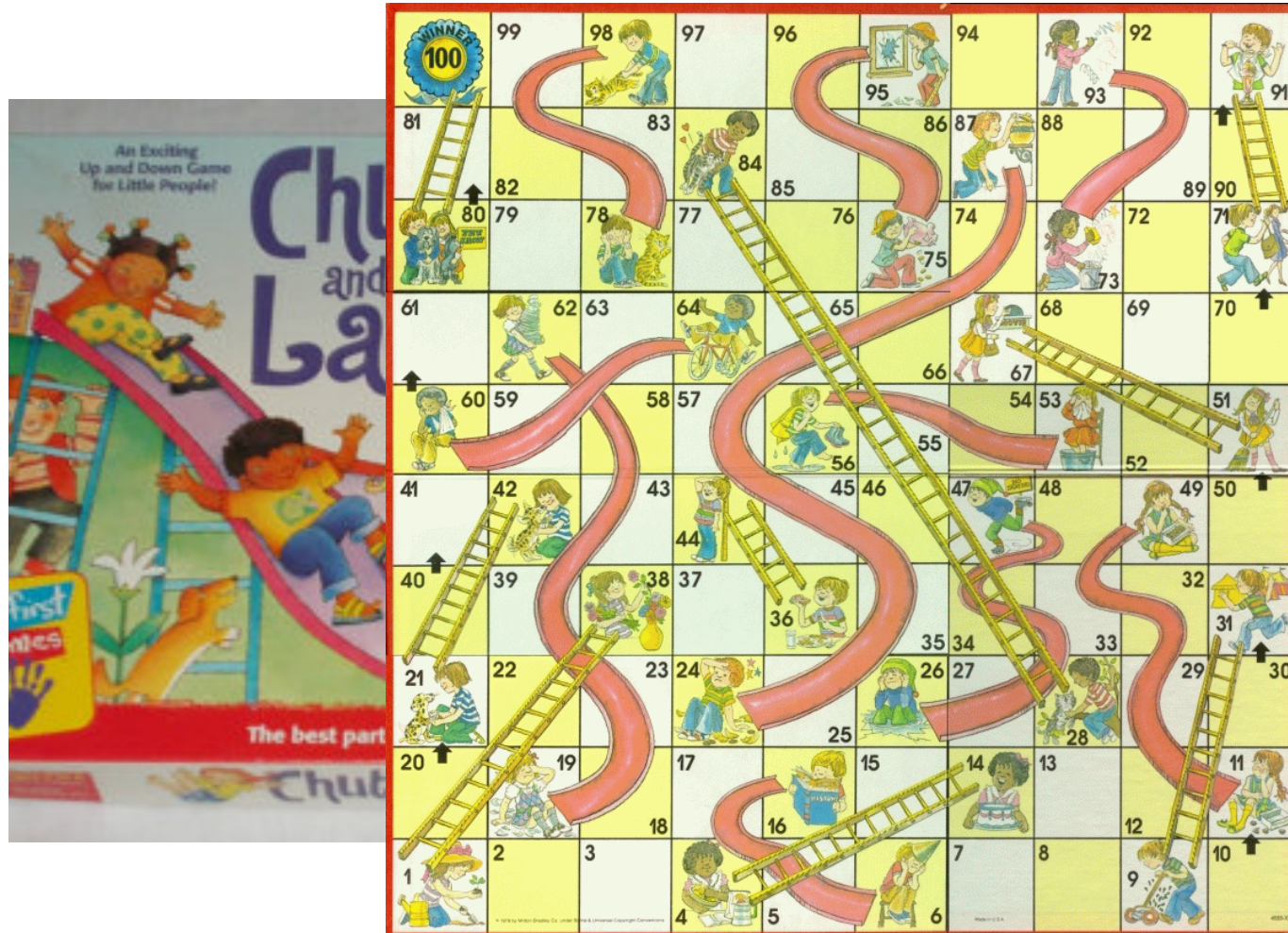


R for Data Science

How analyses are in reality



How analyses are in reality



How analyses are presented



Be complete ...

... but not *too* complete.

P8111HW3

3/8/16, 2:50 PM

```
## [12,] -0.08394047 -0.032693143
## [13,] -0.08403121 -0.034394585
## [14,] -0.08397860 -0.034980378
## [15,] -0.08434735 -0.036156917
## [16,] -0.08435698 -0.036650398
## [17,] -0.08447201 -0.037679892
## [18,] -0.08412698 -0.037754627
## [19,] -0.08435342 -0.038675913
## [20,] -0.08403146 -0.038832864
## [21,] -0.08331837 -0.038555963
## [22,] -0.08311964 -0.038486428
## [23,] -0.08264245 -0.038160749
## [24,] -0.08228691 -0.037783216
## [25,] -0.08166935 -0.037351882
## [26,] -0.08118348 -0.036826247
## [27,] -0.08011525 -0.035707460
## [28,] -0.07967686 -0.035021461
## [29,] -0.07867025 -0.033853880
## [30,] -0.07820231 -0.033233691
## [31,] -0.07719687 -0.032100156
## [32,] -0.07682622 -0.031555885
## [33,] -0.07581413 -0.030493934
## [34,] -0.07555181 -0.029784794
## [35,] -0.07424381 -0.028273179
## [36,] -0.07362121 -0.027558268
## [37,] -0.07244494 -0.026111580
## [38,] -0.07187484 -0.025583699
## [39,] -0.07056949 -0.024287362
## [40,] -0.06983318 -0.023357617
## [41,] -0.06851434 -0.022093530
## [42,] -0.06782302 -0.021520307
## [43,] -0.06691565 -0.020101802
## [44,] -0.06638024 -0.019371473
## [45,] -0.06541473 -0.018089351
## [46,] -0.06504171 -0.017422153
## [47,] -0.06337542 -0.016354797
## [48,] -0.06284136 -0.015832022
## [49,] -0.06196227 -0.014421761
## [50,] -0.06160143 -0.013729093
## [51,] -0.06073027 -0.012330712
## [52,] -0.06035373 -0.011730336
## [53,] -0.05985272 -0.010589271
## [54,] -0.05976229 -0.010116416
## [55,] -0.05931431 -0.009188999
## [56,] -0.05919370 -0.008598205
## [57,] -0.05923929 -0.007870010
## [58,] -0.05884408 -0.007463907
```

Be complete ...

... but not *too* complete.

PB111HW3			PB111HW3			3/6/15, 2:50 PM			3/6/15, 2:50 PM		
##	[12,]	-0.08394047	-0.032693143	##	[59,]	-0.05828045	-0.006427910	##	[59,]	-0.05828045	-0.006427910
##	[13,]	-0.08403121	-0.034394585	##	[60,]	-0.05849033	-0.006043732	##	[60,]	-0.05849033	-0.006043732
##	[14,]	-0.08397860	-0.034980378	##	[61,]	-0.05889288	-0.005276025	##	[61,]	-0.05889288	-0.005276025
##	[15,]	-0.08434735	-0.036156917	##	[62,]	-0.05939320	-0.004944063	##	[62,]	-0.05939320	-0.004944063
##	[16,]	-0.08435698	-0.036650398	##	[63,]	-0.06050816	-0.004206706	##	[63,]	-0.06050816	-0.004206706
##	[17,]	-0.08447201	-0.037679892	##	[64,]	-0.06104112	-0.003886241	##	[64,]	-0.06104112	-0.003886241
##	[18,]	-0.08412698	-0.037754627	##	[65,]	-0.06174906	-0.003648505	##	[65,]	-0.06174906	-0.003648505
##	[19,]	-0.08435342	-0.038675913	##	[66,]	-0.06241750	-0.003800381	##	[66,]	-0.06241750	-0.003800381
##	[20,]	-0.08403146	-0.038832864	##	[67,]	-0.06358717	-0.003662780	##	[67,]	-0.06358717	-0.003662780
##	[21,]	-0.08331837	-0.038555963	##	[68,]	-0.06437141	-0.003635112	##	[68,]	-0.06437141	-0.003635112
##	[22,]	-0.08311964	-0.038486428	##	[69,]	-0.06606900	-0.003829907	##	[69,]	-0.06606900	-0.003829907
##	[23,]	-0.08264245	-0.038160749	##	[70,]	-0.06673738	-0.004057332	##	[70,]	-0.06673738	-0.004057332
##	[24,]	-0.08228691	-0.037783216	##	[71,]	-0.06808891	-0.004816457	##	[71,]	-0.06808891	-0.004816457
##	[25,]	-0.08166935	-0.037351882	##	[72,]	-0.06912238	-0.005185557	##	[72,]	-0.06912238	-0.005185557
##	[26,]	-0.08118348	-0.036826247	##	[73,]	-0.07133348	-0.005956885	##	[73,]	-0.07133348	-0.005956885
##	[27,]	-0.08011525	-0.035707460	##	[74,]	-0.07252225	-0.006506062	##	[74,]	-0.07252225	-0.006506062
##	[28,]	-0.07967686	-0.035021461	##	[75,]	-0.07551133	-0.007708559	##	[75,]	-0.07551133	-0.007708559
##	[29,]	-0.07867025	-0.033853880	##	[76,]	-0.07707811	-0.008382036	##	[76,]	-0.07707811	-0.008382036
##	[30,]	-0.07820231	-0.033233691	##	[77,]	-0.07985765	-0.009874331	##	[77,]	-0.07985765	-0.009874331
##	[31,]	-0.07719687	-0.032100156	##	[78,]	-0.08089292	-0.010706543	##	[78,]	-0.08089292	-0.010706543
##	[32,]	-0.07682622	-0.031555885	##	[79,]	-0.08369384	-0.012351292	##	[79,]	-0.08369384	-0.012351292
##	[33,]	-0.07581413	-0.030493934	##	[80,]	-0.08526338	-0.013185658	##	[80,]	-0.08526338	-0.013185658
##	[34,]	-0.07555181	-0.029784794	##	[81,]	-0.08823390	-0.014956617	##	[81,]	-0.08823390	-0.014956617
##	[35,]	-0.07424381	-0.028273179	##	[82,]	-0.08979044	-0.016127845	##	[82,]	-0.08979044	-0.016127845
##	[36,]	-0.07362121	-0.027558268	##	[83,]	-0.09304215	-0.018502398	##	[83,]	-0.09304215	-0.018502398
##	[37,]	-0.07244494	-0.026111580	##	[84,]	-0.09474251	-0.019530388	##	[84,]	-0.09474251	-0.019530388
##	[38,]	-0.07187484	-0.025583699	##	[85,]	-0.09829321	-0.022059960	##	[85,]	-0.09829321	-0.022059960
##	[39,]	-0.07056949	-0.024287362	##	[86,]	-0.10029588	-0.023170549	##	[86,]	-0.10029588	-0.023170549
##	[40,]	-0.06983318	-0.023357617	##	[87,]	-0.10406055	-0.025920217	##	[87,]	-0.10406055	-0.025920217
##	[41,]	-0.06851434	-0.022093530	##	[88,]	-0.10600856	-0.027101026	##	[88,]	-0.10600856	-0.027101026
##	[42,]	-0.06782302	-0.021520307	##	[89,]	-0.11050742	-0.030160113	##	[89,]	-0.11050742	-0.030160113
##	[43,]	-0.06691565	-0.020101802	##	[90,]	-0.11284952	-0.031782676	##	[90,]	-0.11284952	-0.031782676
##	[44,]	-0.06638024	-0.019371473	##	[91,]	-0.11767979	-0.035136968	##	[91,]	-0.11767979	-0.035136968
##	[45,]	-0.06541473	-0.018089351	##	[92,]	-0.12016788	-0.037081378	##	[92,]	-0.12016788	-0.037081378
##	[46,]	-0.06504171	-0.017422153	##	[93,]	-0.12527376	-0.040924296	##	[93,]	-0.12527376	-0.040924296
##	[47,]	-0.06337542	-0.016354797	##	[94,]	-0.12789947	-0.043164742	##	[94,]	-0.12789947	-0.043164742
##	[48,]	-0.06284136	-0.015832022	##	[95,]	-0.13313524	-0.047185703	##	[95,]	-0.13313524	-0.047185703
##	[49,]	-0.06196227	-0.014421761	##	[96,]	-0.13570119	-0.049292382	##	[96,]	-0.13570119	-0.049292382
##	[50,]	-0.06160143	-0.013729093	##	[97,]	-0.14100031	-0.053871825	##	[97,]	-0.14100031	-0.053871825
##	[51,]	-0.06073027	-0.012330712	##	[98,]	-0.14373346	-0.056295080	##	[98,]	-0.14373346	-0.056295080
##	[52,]	-0.06035373	-0.011733036	##	[99,]	-0.14936673	-0.061355509	##	[99,]	-0.14936673	-0.061355509
##	[53,]	-0.05985272	-0.010589271	##	[100,]	-0.15226672	-0.063875523	##	[100,]	-0.15226672	-0.063875523
##	[54,]	-0.05976229	-0.010116416	##	[101,]	-0.15837583	-0.069434604	##	[101,]	-0.15837583	-0.069434604
##	[55,]	-0.05931431	-0.009188999	##	[102,]	-0.16170942	-0.072208042	##	[102,]	-0.16170942	-0.072208042
##	[56,]	-0.05919370	-0.008598205	##	[103,]	-0.16861004	-0.077481323	##	[103,]	-0.16861004	-0.077481323
##	[57,]	-0.05923929	-0.007870010	##	[104,]	-0.17215587	-0.080216911	##	[104,]	-0.17215587	-0.080216911
##	[58,]	-0.05884408	-0.007463907	##	[105,]	-0.17936529	-0.085934601	##	[105,]	-0.17936529	-0.085934601

Be complete ...

... but not *too* complete.

PB111HW3
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PB111HW3
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3/6/15, 2:50 PM

```
## [12,] -0.08394047 -0.032693143
## [13,] -0.08403121 -0.034394585
## [14,] -0.08397860 -0.034980378
## [15,] -0.08434735 -0.036156917
## [16,] -0.08435698 -0.036650398
## [17,] -0.08447201 -0.037679892
## [18,] -0.08412698 -0.037754627
## [19,] -0.08435342 -0.038675913
## [20,] -0.08403146 -0.038832864
## [21,] -0.08331837 -0.038555963
## [22,] -0.08311964 -0.038486428
## [23,] -0.08264245 -0.038160749
## [24,] -0.08228691 -0.037783216
## [25,] -0.08166935 -0.037351882
## [26,] -0.08118348 -0.036826247
## [27,] -0.08011525 -0.035707460
## [28,] -0.07967686 -0.035021461
## [29,] -0.07867025 -0.033853880
## [30,] -0.07820231 -0.033233691
## [31,] -0.07719687 -0.032100156
## [32,] -0.07682622 -0.031555885
## [33,] -0.07581413 -0.030493934
## [34,] -0.07555181 -0.029784794
## [35,] -0.07424381 -0.028273179
## [36,] -0.07362121 -0.027558268
## [37,] -0.07244494 -0.026111580
## [38,] -0.07187484 -0.025583699
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## [46,] -0.06504171 -0.017422153
## [47,] -0.06337542 -0.016354797
## [48,] -0.06284136 -0.015832022
## [49,] -0.06196227 -0.014421761
## [50,] -0.06160143 -0.013729093
## [51,] -0.06073027 -0.012330712
## [52,] -0.06035373 -0.011730336
## [53,] -0.05985272 -0.010589271
## [54,] -0.05976229 -0.010116416
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## [56,] -0.05919370 -0.008598205
## [57,] -0.05923929 -0.007870010
## [58,] -0.05884408 -0.007463907
```

```
## [59,] -0.05828045 -0.006427910
## [60,] -0.05849033 -0.006043732
## [61,] -0.05889288 -0.005276025
## [62,] -0.05939320 -0.004944063
## [63,] -0.06050816 -0.004206706
## [64,] -0.06104112 -0.003886241
## [65,] -0.06174906 -0.003648505
## [66,] -0.06241750 -0.003800381
## [67,] -0.06358717 -0.003662780
## [68,] -0.06437141 -0.003635112
## [69,] -0.06606900 -0.003829907
## [70,] -0.06673738 -0.004057332
## [71,] -0.06808891 -0.004816457
## [72,] -0.06912238 -0.005185557
## [73,] -0.07133348 -0.005956885
## [74,] -0.07252225 -0.006506062
## [75,] -0.07551133 -0.007708559
## [76,] -0.07707811 -0.008382036
## [77,] -0.07985765 -0.009874331
## [78,] -0.08089292 -0.010706543
## [79,] -0.08369384 -0.012351292
## [80,] -0.08526338 -0.013185658
## [81,] -0.08823390 -0.014956617
## [82,] -0.08979044 -0.016127845
## [83,] -0.09304215 -0.018502398
## [84,] -0.09474251 -0.019530388
## [85,] -0.09829321 -0.022059960
## [86,] -0.10029588 -0.023170549
## [87,] -0.10406055 -0.025920217
## [88,] -0.10600856 -0.027101026
## [89,] -0.11050742 -0.030160113
## [90,] -0.11284952 -0.031782676
## [91,] -0.11767979 -0.035133698
## [92,] -0.12016788 -0.037081378
## [93,] -0.12527376 -0.040924296
## [94,] -0.12789947 -0.043164742
## [95,] -0.13313524 -0.047185703
## [96,] -0.13570119 -0.049292382
## [97,] -0.14100031 -0.053871825
## [98,] -0.14373346 -0.056295080
## [99,] -0.14936673 -0.061355509
## [100,] -0.15226672 -0.063875523
## [101,] -0.15837583 -0.069434604
## [102,] -0.16170942 -0.072208042
## [103,] -0.16861004 -0.077481323
## [104,] -0.17215587 -0.080216911
## [105,] -0.17936529 -0.085934601
```

```
## [200,] -0.78579562 -0.671141895
## [201,] -0.78649970 -0.670452918
## [202,] -0.78675680 -0.669902358
## [203,] -0.78512608 -0.668247942
## [204,] -0.78453474 -0.666857604
## [205,] -0.78443188 -0.662315516
## [206,] -0.78302324 -0.661049715
## [207,] -0.78035282 -0.657166622
## [208,] -0.77942284 -0.654408116
## [209,] -0.77706316 -0.647297818
## [210,] -0.77494542 -0.644028816
## [211,] -0.77142512 -0.637076968
## [212,] -0.76967333 -0.632319394
## [213,] -0.76557589 -0.623728406
## [214,] -0.76276779 -0.619648160
## [215,] -0.75721274 -0.609000925
## [216,] -0.75211326 -0.603117713
## [217,] -0.74700729 -0.589563950
## [218,] -0.74318011 -0.583664200
## [219,] -0.73463678 -0.567231414
## [220,] -0.73067486 -0.559006529
## [221,] -0.72235137 -0.541947505
```

Confidence intervals and the fitted value curve in each bootstrap sample is done above.

Part e)

```
### plot them using ggplot
eachcil<-as.data.frame(eachci)
plot = ggplot() + geom_point(color = "black") +
  geom_line(aes(x=lidar$range, y = UB), color = "blue", linetype = 2) +
  geom_line(aes(x=lidar$range, y = LB), color = "blue", linetype = 2) +
  geom_line(data=eachcil, aes(x=lidar$range, y = V2), color = "green", linetype = 2) +
  geom_line(data=eachcil, aes(x=lidar$range, y = V1), color = "green", linetype = 2)
plot
```

Strike a balance

- This is where practice comes in

R Markdown?

- A “Markdown” language is a *lightweight syntax* that can be easily converted to another format (HTML, PDF, Word)
- R Markdown lets you combine formatted text with code chunks and the results of those chunks



- Having text and code in the same place, and having the combined output be user-friendly, is huge for your workflow

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Time to code!!